TATA 1mg PEOPLEDESK · AI & DATABASE ARCHITECTURE SPECIFICATION

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AI Department Creation Engine & Cloud Database Persistence

Complete architectural and operational blueprint governing **automated HR Department creation via AI**, two-stage machine learning intent training (English + Hinglish), and **real-time Neon Cloud PostgreSQL persistence** for zero-loss operations on Render.

SECTION 1 3-STAGE AI DEPARTMENT & INTENT CLASSIFICATION ENGINE

Whenever an incoming employee email arrives at the HR Flow Inbox, the system executes an automated, deterministic 3-stage AI & NLP routing engine that mimics the enterprise routing models of LEAVE and ATTENDANCE. Admins can dynamically provision any new department (e.g., *Legal, Talent Acquisition, IT Helpdesk, Benefits, BGV*) and immediately train dedicated NLP classifiers without restarting the server.

Stage 1: Department Detection Router	Stage 2: LinearSVC Intent Classifier	Stage 3: Live Verification Studio
Weighted Keyword Scoring: • Strong Phrases: **+3.2** Subject / **+2.2** Body • Weak Keywords: **+1.5** Subject / **+0.8** Body • Threshold: Score **>= 1.5** routes email • Supports English & Workplace Hinglish • Stored in hr_department_keyword_rules	**scikit-learn + TF-IDF Model:** • 400 labeled emails (~65% Eng, 35% Hinglish) • 8 distinct operational intent codes • Generates <dept>_linear_svm.pkl and <dept>_tfidf_vectorizer.pkl • Automatic fallback weights generator • 0ms Hot-Reload into memory	**Real-Time Validation Console:** • Test Subject & Body inputs in UI • Outputs Detected Category & Confidence % • Outputs Predicted Intent Code & Name • Computes SVM Margin & Score • Verifies target SLA & Desk Assignment

SECTION 2 STEP-BY-STEP AI DEPARTMENT PROVISIONING WORKFLOW

STEP	OPERATIONAL ACTION & UI INTERFACE	AI / SYSTEM LOGIC	DATABASE COMMIT
Step 1	Click **+ Add New HR Department** in Admin Desk. Enter Name, Shared Gmail, SLA Hours, Priority, and Description.	Auto-generates DEPT-CODE and normalized category_code. Sets default SLA and routing priority.	Inserts into hr_departments and auto-creates universal sub-department.
Step 2	Click ■ **Copy ChatGPT Keyword Prompt**. Paste ChatGPT's generated Strong & Weak keyword lists into Stage 1.	Generates 20–28 multi-word phrases & 15–22 single keywords in both English & Indian Hinglish.	Inserts into hr_department_keyword_rules with 1.5 routing threshold.
Step 3	Inside ■ **Train AI** drawer, click **Copy ChatGPT Intent Prompt**. Paste the 400-sample JSON into textarea & click **Train**.	Executes train_department_intent_model(). Trains LinearSVC & TF-IDF vectorizer, saves .pkl artifacts, and hot-reloads model.	Saves training_dataset.json, model_metadata.json, and trained weights.
Step 4	Enter sample English/Hinglish Subject & Body in **Stage 3 Tester** and click **Test Live**.	Calls /api/classification/detect-department-and-intent to verify Confidence %, SVM Score, and Margin.	Validates routing readiness for production inbox triage.

SECTION 3 DATABASE ARCHITECTURE & NEON POSTGRESQL CLOUD SYNCHRONIZATION

HR Flow operates a high-performance **Dual-Database Architecture** designed for zero latency and cloud durability. Transactions execute locally against SQLite with microsecond latency, while backend/database/neon_sync.py automatically replicates snapshots to **Neon Serverless PostgreSQL** (AWS ap-southeast-1) to survive ephemeral Render restarts.

TABLE NAME	KEY COLUMNS & DATA TYPES	FUNCTION & REPLICATION SCOPE
hr_departments	`id (INT PK), name (TEXT), dept_code (TEXT), dept_gmail (TEXT), category_scope (TEXT), sla_hours (INT), default_priority (TEXT), description (TEXT)`	Master directory of HR desks. Every created desk dynamically cascades into User Management dropdowns and case routing.
hr_department_keyword_rules	`id (INT PK), department_id (INT FK), category_code (TEXT), keywords_strong (TEXT JSON), keywords_weak (TEXT JSON), scoring_threshold (REAL)`	Stage 1 detection weights. Stores strong phrases (+3.2/+2.2) and weak keywords (+1.5/+0.8) for live email scanning.
hr_department_members	`id (INT PK), department_id (INT FK), member_name (TEXT), email (TEXT), role (TEXT), designation (TEXT), sub_department (TEXT NOT NULL)`	Specialist assignments. Defaulted to 'Universal Desk (General)' to prevent SQL constraint exceptions.
hrflow_sqlite_snapshots ***(Neon PostgreSQL)***	`filename (TEXT PK), data (BYTEA compressed), size_bytes (BIGINT), updated_at (TIMESTAMP)`	Cloud snapshot persistence. Auto-restored on boot (restore_db_from_neon) and debounced auto-backed up on DB commits.

SECTION 4 DYNAMIC USER MANAGEMENT & DUAL-PORT SMTP DELIVERY

1. Real-Time Roster Injection: When a department is created or updated, /api/agent/employees and /api/cases/departments immediately include it. Admins can elect roles and assign specialists to the new department instantly.
2. 4-Tier RBAC Governance: Tier 1 ADMIN (full governance & AI training), Tier 2 HR_MANAGER (escalations & approvals), Tier 3 HR_LEAD (queue assignment & triage), Tier 4 HR_AGENT (workbench case resolution).
3. Dual-Port SMTP Fallback: Outbound notification emails attempt SSL port 465, and automatically fall back to STARTTLS port 587 to guarantee delivery even when cloud hosts throttle SSL handshakes.

SECTION 5 PRODUCTION JSON TRAINING DATASET SCHEMA (STAGE 2 FORMAT)